Matrix functions and stability

Realizing arbitrary polynomial growth exponents by finite matrix families

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because finite families must realize every exponent at every length; community impact connects switched dynamics and asymptotic matrix-product growth.

Resolution — 2026-09-11

Affirmative resolution. Matthew J. Colbrook's complete manuscript, Theorem 1 and its proof in §§2–5 constructs two distinct real matrices for every real $\alpha \geq 0$, with joint spectral radius one and maximal length-k product norm between positive constant multiples of k^α for every integer $k \geq 1$. The matrices and dimension are fixed once the exponent is chosen. Six-dimensional pairs suffice for $0 < \alpha < 1$; every nonnegative rational exponent can be realized with dyadic-rational entries. The proof covers all switching words for the upper bound and every length for the lower bound, including small lengths. It asserts comparability, not convergence of the normalized growth sequence.

The complete original proof passed independent Codex-agent review. Authored TeX · Submission, authorship and verification record. The proof was developed with AI assistance; no external human peer review or formal verification is claimed. The original statement and prior evidence below are retained, and the ratings above are historical. This entry no longer contributes to the open count.

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d\times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k\to\infty} \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

For a compact nonempty $\mathcal{M} \subset \mathbb{R}^{d\times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, define its maximal product norm at length k by

$$g_{\mathcal{M}}(k) = \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2.$$

This problem concerns growth within one family over time. MF-07 instead requests a dimension-dependent bound uniform across families.

Problem statement

For every real $\alpha \geq 0$, do there exist a positive integer d, a finite nonempty $\mathcal{M} \subset \mathbb{R}^{d\times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, and constants $0 < c \leq C < \infty$ such that

$$ck^\alpha \leq g_{\mathcal{M}}(k) \leq Ck^\alpha \qquad \text{for every integer } k \geq 1?$$

Reference and status evidence

Varney and Morris, On marginal growth rates of matrix products, §7, Question 2. Corollary 6.1 realizes exponent 1/3; the all-exponents question remains posed. Searches for the title, authors, and marginal-growth exponents through 2026 located no complete answer. Infinite compact families and subsequence-only lower bounds do not meet the finite-family, every-length target.

Audit — 2026-09-10

Rechecked Varney–Morris, Corollary 6.1, Proposition 3.1, and Question 2: exponent 1/3 is realized and realizable exponents are closed under addition. The all-exponents assertion remains open. Author and marginal-growth-exponent searches found no complete finite-family construction or obstruction.